

MCS 320 Project One : Interval Arithmetic for Accurate Results due Wednesday 23 September 2026 at 1pm

The goal of this project is to apply interval arithmetic to evaluate expressions accurately:

1. Evaluating expressions at an interval returns an interval. The center of the interval approximates the value of expression. The diameter of the interval on return gives an upper bound on the error of the computed value.
2. Given a desired accuracy, we increase the precision of the interval arithmetic gradually till the upper bound of the error is sufficiently small.

Below we illustrate an interval arithmetic computation with 8 bits of precision, evaluating $1/(1/10)$. Because $1/10$ has no finite binary expansion, the upper bound on the error of the evaluation of $1/(1/10)$ is 1.3×10^{-2} as computed by the statements below:

```
RIF8 = RealIntervalField(prec=8)
onetenth = RIF8(1/10)
ten = 1/onetenth
ten.diameter()
```

The bracket style representation of `ten` is obtained via `ten.str(style='brackets')`.

It is very important to know that

$$1/\text{RIF8}(1/10) \neq \text{RIF8}(1/(1/10))$$

as the left hand side is the evaluation of $1/(1/10)$ with 8-bit interval arithmetic, while the right hand side is the representation of 10 (computed exactly in rational arithmetic) as an 8-bit precision interval, which is also exact.

If we are satisfied with 3 decimal places of accuracy, then 8-bit floating-point arithmetic will do. Otherwise, for a given desired accuracy, we gradually double the precision under the upper bound for the error is sufficiently small.

1. The Accuracy of Evaluating an Expression

Consider the expression X below

$$X = \left(\frac{27/10 - e}{\pi - (\sqrt{2} + \sqrt{3})} \right)^{67/16}.$$

Assignment One.

1. Evaluate the expression for X with 16-bit precision interval arithmetic.
What is the upper bound on the error?
2. How many bits of precision will be required if we want to know the value of X with 8 decimal places of accuracy?
Gradually double the precision, stopping when the upper bound on the error is less than 5×10^{-8} .
Report the bound on the error in each step.
3. Compare the centers of the intervals of the last two approximations.
How many decimal places are in agreement?

2. The Area of a Thin Triangle

To compute the area A of a triangle with sides a , b , and c , we evaluate

$$a = 7; \quad b = 7/2 + 3 \times 2^{-111}; \quad c = b$$
$$s = (a + b + c)/2; \quad A = \sqrt{s(s-a)(s-b)(s-c)}$$

Assignment Two.

1. Evaluate the formulas with 8-bit precision interval arithmetic.
What is the upper bound for the error on s ? How accurate is A ?
2. If we want the area A to be accurate with 8 decimal places, then how many bits in the precision of the interval arithmetic are needed?
Gradually double the precision, stopping when the upper bound on the error is less than 5×10^{-8} .
What is the relation between the bound on the error of s and the error on A ?
3. Compare the centers of the last two approximations and report the number of decimal places that agree between those centers.
If the accuracy of the approximation for A is less than 8 decimal places, then continue doubling the precision till it is.

3. The Quadratic Formula

Consider the application of the quadratic formula to compute the roots r_1 and r_2 of $3x^2 + 100x + 2$, evaluating

$$a = 3; \quad b = 100; \quad c = 2; \quad d = \sqrt{b^2 - 4ac}$$

$$r_1 = (-b + d)/(2a); \quad r_2 = (-b - d)/(2a)$$

Assignment Three.

1. Evaluate the formulas for the discriminant d and for r_1 , r_2 with 8-bit precision interval arithmetic.
Report the upper bound for the error on d , r_1 , and r_2 .
Can you predict the upper bound for the error on the roots via the error on d ?
2. If we want the roots to be accurate with 8 decimal places, then how many bits in the precision of the interval arithmetic are needed?
Gradually double the precision, stopping when the upper bound on the error is less than 5×10^{-8} .
What is the relation between the bound on the error of d and the largest error on the roots?
3. Compare the centers of the last two approximations and report the number of decimal places that agree between those centers.

4. the Deadline is Wednesday 23 September 2026 at 1pm

Upload your solution to gradescope before 1pm on Wednesday 23 September 2026.

The solution consists of one single notebook, organized according to the assignments. Mark the start of each solution to an assignment using a heading in a cell. Your notebook should run from top to bottom as a program without errors. Apply proper formatting in your notebook so it reads like a technical report if you would print it. Document the execution cells with complete sentences, properly formatted in markdown cells.

You may (not must) work in pairs for this project. A pair consists of two, not three or more. If you decide to work in a pair, then you must send an email with the name of your partner and with the email address of your partner in the copy of the email to janv@uic.edu, before 5pm on Wednesday 16 September. If working in a pair, then only one Jupyter notebook should be submitted.

If you have questions or difficulties with the project, feel free to ask questions during office hour.